

FPGAs for The Internet of Things



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Billions of devices are expected to be connected wirelessly by 2020. In this emerging era of connected devices, machines need not only be secure but also need to be secure at device, design and system levels. The Internet of Things (IoT) requires diverse technology and specialised skill areas such as specialised hardware and sensor development, along with sophisticated real-time embedded firmware, cloud applications and Big Data analytics for massive real-time data into usable information, delivery of data to human-scale and human-usable platforms, particularly sophisticated smartphone apps. The IoT is revealing an important need in technology, that is, programmable hardware and I/O.

The IoT will soon be driven by field-programmable gate array (FPGA)-like devices, because these devices can interface with the outside world very easily and provide lowest power, lowest latency and best determinism. The IoT would interface with temperature, pressure, position, acceleration, analogue-to-digital converters (ADCs), digital-to-analogue converters (DACs), current and voltage, among others. Arduino and Raspberry Pi could also be used.

An FPGA can be considered a programmable special-purpose processor

as it can handle signals at its input pins, process these and drive signals on its output pins.

The above system is very deterministic. An FPGA can interact with memory and storage devices through serialiser/deserialiser interfaces (SERDES), which also allow for Ethernet, serial or Bluetooth communication. An FPGA can, for example, take an HTTP request packet received from a wireless Ethernet component, decode its request, fetch information from memory and return the requested result back through the Ethernet device.

The FPGA could be coupled with an ARM processor to leverage higher-level software functions such as Web servers or security packages, if higher level of processing is required. The key consideration is the programmable aspect of an FPGA. In a typical development cycle, a supplier development kit is employed to configure the FPGA, while a printed circuit board (PCB) is developed with specific sensor/communication/display components, as required.

The IoT challenges

The IoT challenges include security, privacy, unauthorised access, malicious control and denial of service. A hardware-first approach with respect to security and implementation of necessary functionality on the systems on chip (SoC) level is vital for

Fig. 1: The IoT comprises at least three layers, each with its own medium and protocols

